# Austin A. MBaye

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*Research Interests: Topological Data Analysis · Computational Biology · Time Series Analysis · Computer Vision*

## Education

**Ph.D. in Mathematics · Northeastern University** *Aug 2024 – Present*
*Boston, MA*

- Advisor: Dr. Jose Perea
- Lakshmibai Venkatramani Fellowship

**B.A. in Mathematics (Minor: Education) · Vassar College** *Aug 2020 – May 2024*
*Poughkeepsie, NY*

- Advisor: Dr. Adam Lowrance
- Departmental Honors in Mathematics · Sigma Xi · GPA: 3.82/4.0

## Research Experience

**Visiting Researcher – Max Planck Institute of Molecular Cell Biology and Genetics (MPI-CBG)** *Jun 2026 – Jul 2026*
*Dresden, Germany*

- Under Dr. Heather Harrington, and Dr. Stephan Grill, applied **Chiral Multiparameter Persistence (CaMPers)** to analyze actomyosin cortical flow fields and quail nematic fields, quantifying symmetry breaking and directional bias in embryonic cell dynamics.

**Graduate Research Assistant – Perea Lab, Northeastern University** *Sep 2024 – Present*
*Boston, MA*

- Developed **Chiral Multiparameter Persistence (CaMPers)**, a novel TDA framework for detecting and quantifying chirality in vector fields on Riemannian manifolds; proved stability theorems and introduced chiral Euler Characteristic Curves (cECC).
- Developed **AQSM_SW1PerS**, a persistent homology pipeline for automated quantification of stereotypical motor movements in autism from multimodal sensor data; validated on real-world behavioral datasets.

**Undergraduate Researcher – MSRI Undergraduate Program (MSRI-UP)** *Jun 2023 – Jul 2023*
*Berkeley, CA*

- Selected as one of 18 students nationwide for a competitive NSF-funded REU at the Mathematical Sciences Research Institute.
- Co-designed a persistent cohomology algorithm using cup products to detect quasi-periodicity in univariate time series; implemented the full pipeline in Python.
- Presented findings at MSRI and co-authored a technical report distributed to program participants.

## Publications and Preprints

*Publications*

1. **MBaye, A.A.**, Perea, J.A., Tralie, C.J., & Goodwin, M.S. *Automated quantification of stereotypical motor movements in autism using persistent homology. Sci Rep*, 2026. https://doi.org/10.1038/s41598-026-60095-8
   *Software*: `AQSM_SW1PerS` *Dataset*: figshare doi:10.6084/m9.figshare.30100669.v1

## Research Presentations

*Invited Talks*

- *Automated Quantification of Stereotypical Motor Movements in Autism.* 5th International Conference on Neuroscience & Psychiatry, Paris, France. Nov. 2025. [link]
- *Automated Quantification of Stereotypical Motor Movements in Autism.* Mathematics Colloquium, Vassar College, Poughkeepsie, NY. Jan. 2024. [link]

*Contributed Talks*

- *Chiral Multiparameter Persistence.* Joint Mathematics Meetings, Chicago, IL. Jan. 2027. [link]

- *Automated Quantification of Stereotypical Motor Movements in Autism.* Joint Mathematics Meetings, Chicago, IL. Jan. 2027. [link]
- *Chiral Multiparameter Persistence.* AMS Fall Western Sectional Meeting, Tempe, AZ. Nov. 2026. [link]
- *Automated Quantification of Stereotypical Motor Movements in Autism.* AMS Spring Eastern Sectional Meeting, Hartford, CT. Apr. 2025. [link]
- *Using Persistent Cup Products for Quasi-Periodicity Detection.* Joint Mathematics Meetings, San Francisco, CA. Jan. 2024. [link]
- *Using Persistent Cup Products for Quasi-Periodicity Detection.* MSRI-UP Final Presentation, Berkeley, CA. Jul. 2023. [link]

*Posters*

- *Using Persistent Cup Products for Quasi-Periodicity Detection.* SACNAS National Diversity in STEM Conference, Portland, OR. Oct. 2023.

## Software

- `CaMPers` – Python/C++ package for computing chiral Euler characteristic curves from flow fields. (In development)
- `AQSM_SW1PerS` – Python package implementing a persistent homology pipeline for automated quantification of stereotypical motor movements from behavioral video and accelerometer data.

## Awards and Fellowships

- **Lakshmibai Venkatramani Fellowship** for Mathematics Graduate Students – Northeastern University (2026)
- **MSRI-UP Fellowship** – Mathematical Sciences Research Institute (2023)
- **Sigma Xi** – Scientific Research Honor Society, inducted Vassar College chapter (2024)
- **Departmental Honors in Mathematics** – Vassar College (2024)